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Markov Chain Monte Carlo Methods for Inference in Hidden Markov Models

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1 Abstract

Hidden Markov Models (HMM's) are Markov processes that are only observable through noise. We differentiate between a hidden Markov process $\{X_k\}_{k\geq 0}$ and an the observable states $\{Y_k\}_{k\geq 0}$ that conditionally depend on the hidden states. HMM's have a wide range of applications from speech processing to stock price modeling to only name a few. Making inferences on the distribution of the hidden states given the observable states is called smoothing. Smoothing may prove difficult in cases where exact algorithms or closed-form formulas are not available. In those scenarios simulation based techniques can be used to sample the hidden states conditionally on the observable states. This thesis presents the Accept-Reject algorithm, the Metropolis-Hastings algorithm and Gibbs sampling as three viable options for approximating the hidden states' distribution. To illustrate the different algorithms and their behaviour we conducted a number of numerical simulations. Furthermore, a simulation of a target distribution of a HMM process modelling stock price changes is used to illustrate the Gibbs sampling algorithm.

2 Introduction

Many real life phenomena can be described by a mathematical model called a *Markov process* or *Markov chain*. A Markov chain describes a sequence of states and the corresponding probabilities of transitioning from one state to another, thus making it a stochastic model. The main feature of a Markov chain is the so-called *Markov property*, i.e. the probability of transitioning from the current state to the next one, only depends on the current state. Thus, the whole sequence of states before the present one, does not influence future transitions. This feature is often called "memory-less" as predictions about future states can be made solely on the basis of the present state. The Markov property makes it easier to describe a stochastic model. If future states were dependent on several or all prior states, a description of the process would quickly become difficult. Due to the Markov property the description of a Markov chain comes down to an initial distribution of states and the transition probabilities between states. An example phenomenon that can be modelled with a Markov chain is the stock market. In a simplified version of stock prices, one could consider the stock price change compared to the day before. Either the stock price stagnates, increases or decreases. Given one of the three options for the current state, we could come up with transition probabilities for price change in the next day. Arguably, stock price changes do not only depend on the price change the day before and it can be argued that there are a range of economic factors influencing stock prices. However, for a simple model of stock price development we will consider this model as an example Markov chain. A graphical representation of a Markov chain with the three states price stagnation, price increase and price decrease is shown in Figure 2. The probabilities on the arrows between the states are the transition probabilities. Though very simple, we can model stock price changes with this model.

Building up from the notion of a Markov chain, we will now introduce *Hidden Markov models* (HMM). In our previous example the different states were accessible in a sense that we knew in which state we are in. Now, an HMM is a model where there is a Markov chain that can only be observed through some noise or inaccuracy. Suppose $\{X_k\}_{k\geq 0}$ is a Markov chain on an arbitrary state space with integer index k. This Markov chain is hidden in the sense, that it is not observable. Observable on the other hand is another stochastic process $\{Y_k\}_{k\geq 0}$, which is related to $\{X_k\}$ in the sense that X_k governs the distribution of Y_k. A graphical representation of such a process is given below in Figure 2. The nodes represent to the random variables and the arrows represent the structural dependence of the variables. In chapter 3 there will be a formal introduction to concepts that have so far been introduced with an emphasis on intuitive explanations.

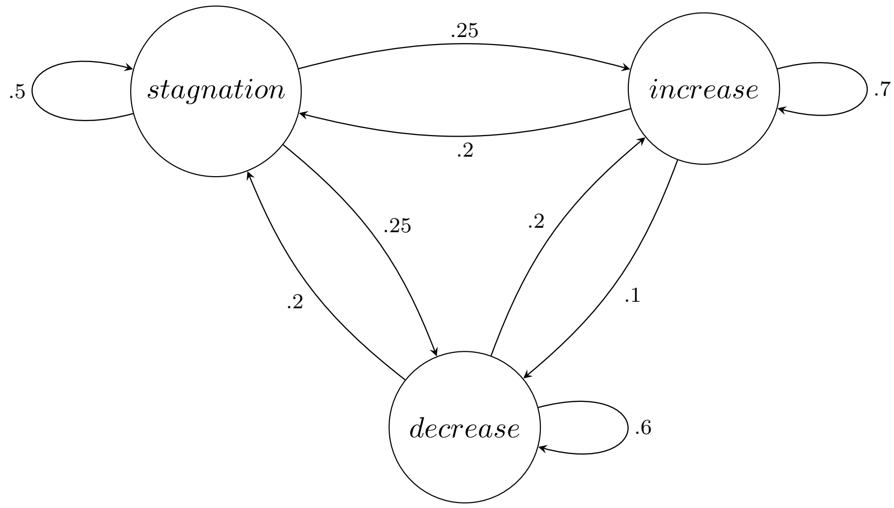

Figure 1: A diagram of the stock price Markov chain with price stagnation, price increase and price decrease as states. The arrows represent the transition probabilities.

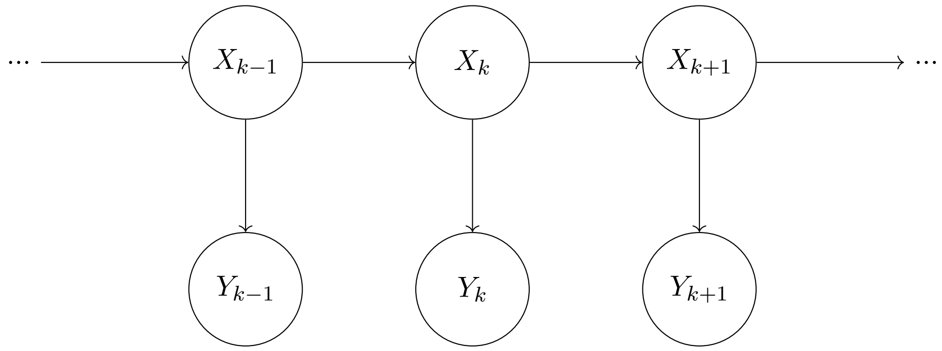

Figure 2: Graphical representation of an HMM with the hidden chain $\{X_k\}$ and the observable process $\{Y_k\}$.

A basic example of an HMM is to choose $\{X_k\}$ as the the symmetric nearest neighbour random walk on $\mathbb{Z}$ with transition probability $1/2$ in both directions. For $\{Y_k\}$ consider a normal distribution with a mean of X_k. Then our observable process is distributed $Y_k \sim \mathcal{N}(X_k, \sigma)$. A graphical representation is given in Figure 3.

As HMM's have the unique structure of hidden states we are interested in ways in which we can make inferences about the hidden states based on the observed sequence. The conditional distribution of a state X_k or several states $X_l, ..., X_k$ given the observations $Y_0, ..., Y_n$ is called *smoothing.* Finding and evaluating this distribution is not always easy or even analytically possible. In cases, where exact algorithm may not be applicable,

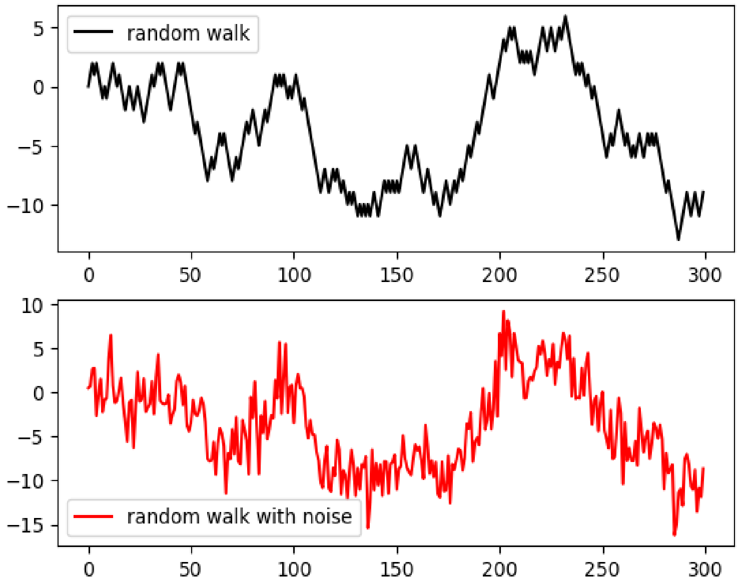

Figure 3: Top: a symmetric random walk on $\mathbb{Z}$ with $1/2$ as transition probability as the hidden chain. Bottom: The observable process $Y_k \sim \mathcal{N}(X_k, 2)$.

one can use simulation based techniques. Methods that will be covered here include the *Accept-Reject algorithm*, the *Metropolis-Hastings algorithm* and *Gibbs sampling*.

A more advanced example for an HMM that we will come back to later on, is the *stochastic volatility model*. Similar to our example Markov chain about stock prices changes, we will again look at prices of financial assets, however this time we are interested in the distributions of these prices. Suppose S_k is the price of a stock at time k. Instead of looking at the actual prices we will take the log return into consideration i.e. $log(S_k - S_{k-1})$, which describes the relative change of the stock price over time. In our model the index k will be the day. Looking at stock price returns, the variance of the returns changes over time. Attempts to model data on returns therefore have a multiplicative form: $Y_k = \sigma_k V_k$, where $\{V_k\}_{k\geq 0}$ is an i.i.d. sequence and the so called volatility process $\{\sigma_k\}_{k\geq 0}$ is a non-negative stochastic process. The term *volatility* is due to the fact that in econometrics the word volatility is used instead of standard deviation. In addition V_k and σ_k are independent for all k. A natural assumption on V_k is that it is symmetric with a mean of 0. Thus, V_k is modelling the direction of price changes and the order of magnitude of the change is modelled by σ_k. A more elaborate approach is a model where the variance or volatility follows some latent stochastic process. Those models are in fact called stochastic volatility models. The standard stochastic volatility model in discrete time looks as follows [Hull and White, 1987, Jacquier et al., 2002]:

$$X_{k+1} = \phi X_k + \sigma U_k, \qquad U_k \sim N(0,1),$$
$$Y_k = \beta \exp(X_k/2) V_k, \qquad V_k \sim N(0,1).$$

Here the observations $\{Y_k\}_{k\geq 0}$ are the log-returns and tconsideration i.e.he latent process $\{X_k\}_{k\geq 0}$ is the log-volatility. The parameter β is a constant scaling factor, ϕ is the persistence or memory of the volatility and σ is the variance of the log-volatility. Given the parameters β, ϕ and σ the goal is to estimate the latent process $\{X_k\}_{k\geq 0}$. Later on, when we have the necessary concepts and algorithms to further analyse this model we will come back to this example.

After having introduced some major concepts intuitively, we will continue with a Preliminaries section, where major concepts such as Hidden Markov models and other useful definitions and statements are introduced rigorously. In Chapter 4, consisting of 3 main subsections the 3 main algorithms will be introduced. First the Accept-Reject algorithm is explained and proven to be correct. Then, the Metropolis-Hastings algorithm is introduced. More specifically we will look at the independent and random walk implementation of the algorithm. As the final major algorithm the Gibbs sampler is introduced. A variation of Gibbs sampling, the so called slice sampler will be introduced and used to approximate the hidden states' distribution of a HMM example about stock price changes. All simulations were implemented in the programming language Python. Lastly, in the conclusion section the algorithms are summarised and discussed with regard to their use and practicability.

3 Preliminaries

This section will introduce some already mentioned concepts in a rigorous way. Note that for the purpose of this work and in order to reduce some complexity, we will only consider the discrete time case, i.e. the chain is a countable infinite sequence and the time steps the chain moves in are discrete: $\{X_k\}_{k\in\mathbb{N}}$.

A first step towards defining a Markov chain is to introduce the concept of a transition kernel. Intuitively, this kernel is the transition probability of transitioning from one state to the next. More general without an immediate relation to a Markov chain we define a transition kernel as follows.

Definition 3.1 (Transition Kernel). *Let $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$ be two measurable spaces. A transition kernel from $(X, \mathcal{X})$ to $(Y, \mathcal{Y})$ is a function*
$Q : X \times \mathcal{Y} \to [0, \infty]$ that satisfies the following conditions:
(i) for all $x \in X$, $Q(x, \cdot)$ is a probability measure on $(Y, \mathcal{Y})$;
(ii) for all $A \in \mathcal{Y}$, the function $x \mapsto Q(x, A)$ is measurable.

In addition, a transition kernel Q is said to admit a density with respect to a measure μ on Y, if there exists a non-negative function $q : X \times Y \to [0, \infty]$, measurable with respect to the product σ-field $\mathcal{X} \otimes \mathcal{Y}$, such that

$$Q(x, A) = \int_A q(x, y)\mu(\mathrm{d}y),\ A \in \mathcal{Y}$$

The function q is then referred to as a transition density function.
In the case where X and Y are countable sets, we write $Q(x, y)$ for $Q(x, \{y\})$.

In order to describe the transitioning of states in a Markov chain we will need the concept of a filtration. A filtration $(\mathcal{F}_k)_{k\in\mathbb{N}}$ on $\mathcal{F}$ is an increasing family of σ-algebras $\mathcal{F}_0 \subset \mathcal{F}_1 \subset ... \subset \mathcal{F}$. A stochastic process $\{X_k\}$ is adapted to a filtration $\mathcal{F}$, if X_k is $\mathcal{F}_k$-measurable for all $k \geq 0$. Intuitively, filtrations are a model for how information flows. $\mathcal{F}_k$ is the information that is available at time k. Thus, for an adapted stochastic process at time k we know the values for $(X_0, ..., X_k)$. With this in mind we will introduce the formal definition of a Markov chain.

Definition 3.2 (Markov Chain). *Let $(\Omega, \mathcal{S}, \mathcal{F}, P)$ be a filtered probability space and let Q be a Markov transition kernel on a measurable space $(X, \mathcal{X})$. An X-valued stochastic process $\{X_k\}_{k\geq 0}$ is said to be a Markov chain under P, with respect to the filtration $\mathcal{F}$ and with the transition kernel Q, if it is $\mathcal{F}$-adapted and for all $k \geq 0$ and $A \in \mathcal{X}$,*

$$P(X_{k+1} \in A \mid \mathcal{F}_k) = Q(X_k, A). \tag{3.1}$$

The distribution ν is called the initial distribution of the chain, and X is called the state space. Notice that equation (3.1) conveys the before mentioned Markov property. $P(X_{k+1} \in A \mid \mathcal{F}_k)$ is the probability of $X_{k+1} \in A$ given all the information available up to time k. $Q(X_k, A)$ on the other hand is the transition kernel from state X_k to transition to a state in A. Thus, we could add intermediate steps to clarify the property in (3.1): $P(X_{k+1} \in A \mid \mathcal{F}_k) = P(X_{k+1} \in A \mid X_0, ..., X_k) = P(X_{k+1} \in A \mid X_k) = Q(X_k, A)$.

Defining a hidden Markov model relies on the same concepts that were underlying the normal Markov model, i.e. we have a pair of transition kernels Q and G. Where Q is the kernel for the hidden Markov chain on X and G the transition kernel for from the hidden state space X to the observable state space Y. Together these two kernel can be used to compose a transition kernel on the product space $(X \times Y, \mathcal{X} \otimes \mathcal{Y})$.

Definition 3.3 (Hidden Markov Model)**.** *Let $(X, \mathcal{X})$ and $(Y, \mathcal{Y})$ be two measurable spaces and let Q denote a Markov transition kernel on $(X, \mathcal{X})$ and G a transition kernel from $(X, \mathcal{X})$ to $(Y, \mathcal{Y})$. The Markov transition kernel defined on the product space $(X \times Y, \mathcal{X} \otimes \mathcal{Y})$ is then*

$$T[(x, y), C] = \int\int_C Q(x, \mathrm{d}x')G(x', \mathrm{d}y'), \qquad for\ (x, y) \in X \times Y, C \in \mathcal{X} \otimes \mathcal{Y}. \tag{3.2}$$

The Markov chain $\{X_k, Y_k\}_{k \geq 0}$ with Markov transition kernel T and initial distribution ν, where ν is a probability measure on $(X, \mathcal{X})$, is called a hidden Markov Model.

Note that here we write $Q(x, dx')$ and $G(x', dy')$ as the transition kernels are given inside of an integral and the set to be integrated over is specified on the integral. Even though ν is a measure on X and not on $X \times Y$, we will still use ν as initial distribution, as the distribution of the law of $\{X_k, X_k\}_{k \geq 1}$ only depends on the marginal distribution of X_0 and not on (X_0, Y_0). In cases where the model is dominated in the sense that there is a probability measure μ on $(Y, \mathcal{Y})$ and λ on $(X, \mathcal{X})$ such that G and Q are absolutely continuous with respect to the corresponding measure, the transition kernel admits the density function: $t[(x, y), (x', y')] = q(x, x')g(x', y')$. Note that, only considering this definition, the hidden aspect of the hidden Markov model is not justified. Only when $\{X_k\}_{k \geq 0}$ are not observable does the term hidden make sense with the definition. That this definition of a hidden Markov model is consistent with the intuition of conditional independence of the states (i.e. Y_k only conditionally depends on X_k) is shown in Proposition 2.2.4. and Corollary 2.2.5. in the book by Cappé, Moulines and Rydén [Cappe et al., 2005].

We will now introduce some definitions and a theorem regarding Markov chains that will later prove useful. Note that the following definitions are only concerning Markov chains with a discrete state space. Even though generally, the Markov chains we will look at do not necessarily have a discrete state space, for the purposes of the simulations we will consider in the next chapters, we will only need results for a discrete state space. We start with the definition of an *invariant measure.*

Definition 3.4 (Invariant measure)**.** *Let K be a transition kernel and π a probability measure. Then π is said to be invariant or stationary probability measure for the kernel K and the associated chain if*

$$\pi(B) = \int_X K(x, B)\pi(\mathrm{d}x), \forall B \in \mathcal{X}.$$

A chain is said to be stationary if it has a stationary probability measure and $X_0 \sim \pi$. Thus, the marginal distribution of $\{X_k\}_{k\geq 0}$ is independent of k as for all integers l and p, $X_l \sim \pi$ and $X_p \sim \pi$.

Definition 3.5 (Reversible Markov chain)**.** *A stationary Markov chain $\{X_k\}_{k\geq 0}$ with transition kernel K is reversible, if there exists a distribution π such that the following holds:*

$$\pi(x)K(x, x') = \pi(x')K(x', x) \tag{3.3}$$

Equation (3.3) is called the *detailed balance equation.* Intuitively this means that the chain has an equilibrium between states of the chain. The probability of being in state x and moving form x to x' is the same as being in state x' and moving from x' to x.

As proven in the next theorem, the detailed balance condition is a sufficient condition for π to be a stationary measure w.r.t. K.

Theorem 3.6. *Let K be a transition kernel for a Markov chain that satisfies the detailed balance equation with the probability density π, then*
(i) π is the stationary density of the chain and
(ii) the chain is reversible.

Proof. For $B \in \mathcal{X}$

$$\int_X K(x,B)\pi(x)\mathrm{d}x = \int_X \int_B K(x,x')\pi(x)\mathrm{d}x'\mathrm{d}x = \int_X \int_B K(x',x)\pi(x')\mathrm{d}x'\mathrm{d}x$$
$$= \int_B \left(\int_X K(x',x)\mathrm{d}x \right) \pi(x')\mathrm{d}x' = \int_B \pi(x')\mathrm{d}x'.$$

The last equation holds as $\int K(x',x)\mathrm{d}x = 1$. Thus, π is the invariant density of the chain. Statement (ii) follows from the fact that the detailed balance equation holds for π and that by definition 3.5 the chain is then reversible. ■

As mentioned before, based on the observations $Y_0, ..., Y_n$ we would like to infer information on the unobserved state sequence $X_0, ..., X_n$. We define this process called smoothing as a transition kernel from our state space of observations to the state space of the unobservable states. From now on we will use an index notation such as $X_{k:l}$ as shorthand notation for $X_k, ..., X_l$. In addition we will use the notation $\mathcal{F}_b(X)$ for the set of all bounded and measurable functions on X.

Definition 3.7 (Smoothing). *Let k, l, and n be positive indices with $l \geq k$. Denote by $\phi_{\nu,k:l|n}$ the conditional distribution of $X_{k:l}$ given $Y_{0:n}$, that is*

(a) $\phi_{\nu,k:l|n}$ is a transition kernel from $Y^{(n+1)}$ to X^{l-k+1}:
• for any given set $A \in \mathcal{X}^{\otimes(l-k+1)}$, $y_{0:n} \mapsto \phi_{\nu,k:l|n}(y_{0:n}, A)$ is a $\mathcal{Y}^{\otimes(n+1)}$-measurable function,
• for any given sub-sequence $y_{0:n}$, $A \mapsto \phi_{\nu,k:l|n}(y_{0:n}, A)$ is a probability distribution on $(X^{l-k+1}, \mathcal{X}^{\otimes(l-k+1)})$.

(b) $\phi_{\nu,k:l|n}$ satisfies, for any function $f \in \mathcal{F}_b(X^{l-k+1})$,

$$E_\nu[f(X_{k:l})|Y_{0:n}] = \int ... \int f(x_{k:l})\phi_{\nu,k:l|n}(Y_{0:n}, x_{k:l}), \tag{3.4}$$

where the equality holds P_ν-almost surely.

The case $\phi_{\nu,k|n}$ for $n \geq k \geq 0$ of only considering the conditional distribution of X_k given all observable states is called marginal smoothing.

We will now give a proposition with which it will be possible to define smoothing kernels in terms of HMM objects. First we introduce a new concept, the *likelihood* of the observations.

Definition 3.8 (Likelihood)**.** *The likelihood of the observation is the probability density function of* $Y_0, Y_1, ..., Y_n$*, with respect to* μ_n*(denoting the product distribution* $\mu^{\otimes(n+1)}$*) defined, for all* $(y_0, ..., y_n) \in Y^{n+1}$*, by*

$$L_{\nu,n}(y_0, ..., y_n) = \int ... \int \nu(\mathrm{d}x_0) g(x_0, y_0) Q(x_0, \mathrm{d}x_1) g(x_1, y_1) \cdots Q(x_{n-1}, \mathrm{d}x_n) g(x_n, y_n)$$

With this, we are able to formulate the following proposition that will give us a useful form for our smoothing kernel $\phi_{\nu,0:n|n}(y_{0:n}, x_{0:n})$.
Before we continue with the next proposition we will have to introduce some notation. Transition kernels can operate on functions, i.e. if we have a kernel Q, a real measurable function f on Y, we define $Q(x, f) := \int f(y) Q(x, \mathrm{d}y)$, for $x \in X$. For $A \in \mathcal{Y}$ the following notations are equivalent $Q(x, A) = Q(x, \mathbf{1}_A)$, where $\mathbf{1}$ is the indicator function.

Proposition 3.9. *Let* n *be a positive integer and* $y_{0:n} \in Y^{n+1}$ *a sub-sequence such that* $L_{\nu,n}(y_{0:n}) > 0$*. The joint smoothing distribution* $\phi_{\nu,0:n|n}$ *then satisfies*

$$\phi_{\nu,0:n|n}(y_{0:n}, f) = L_{\nu,n}(y_{0:n})^{-1} \int ... \int f(x_{0:n}) \nu(\mathrm{d}x_0) g(x_0, y_0) \prod_{k=1}^{n} Q(x_{k-1} \mathrm{d}x_k) g(x_k, y_k) \tag{3.5}$$

for all functions $f \in \mathcal{F}_b(X^{n+1})$*.*

Proposition 3.9. will not be proved here, but the proof can be found in the book by Cappé, Moulines and Rydén as the proof for Proposition 3.1.4. in [Cappe et al., 2005].

4 Markov Chain Monte Carlo Methods

4.1 The Accept-Reject Algorithm

For certain probability distributions such as the Gaussian, or Gamma distribution, there are efficient sampling techniques such as the *inverse transformation*, which we will briefly explain in the following paragraph.

Recursive random number generators on a computer are able to generate sequences of numbers that are approximately uniformly distributed on $1, ..., m$. These numbers are never random in a strict sense, but rather pseudorandom as they are deterministically generated by an algorithm, which is designed in a way to produce numbers that are approximately random. If we choose $m >> 1$ so that $\frac{1}{m}$ is close to the machine precision and we set $X \sim U([1, ..., m])$, then the random variable $Y \sim \frac{X}{m}$ is approximately $U([0, 1])$ distributed. Now with a way to simulate a uniform distribution we can simulate other distributions with the help of the general distribution function. Suppose $G(p) = inf\{x \in \mathbb{R} : F(x) \geq p\}\ for\ p \in (0, 1)$ is the general distribution function for a cumulative distribution function F. As F is monotonic and right-continuous, $G(p) \leq x \Leftrightarrow p \leq F(x)$. Now if we set $Y \sim U([0, 1])$ and $Z = Q(Y)$:

$$P(Z \leq x) = P(Q(Y) \leq x) = P(Y \leq F(x)) = \int_0^{F(x)} \mathbf{1}_{[0,1])}(y)\mathrm{d}y = F(x). \tag{4.1}$$

Thus, Z is a random variable with a distribution function F. Computationally, this process involves computing the cumulative distribution function and than inverting that function for the general distribution function. In the case of continuous distributions this amounts to integrating the probability density function. This may prove difficult for distributions where this integration is impossible to do analytically. If the goal is to provide i.i.d. samples from an arbitrary distribution, there exist only a limited number of general approaches. One of them is the Accept-Reject algorithm, which will now be presented in greater detail.

The fundamental idea underlying the Accept-Reject algorithm is that we can rewrite a density f as

$$f(x) = \int_0^{f(x)} \mathrm{d}u. \tag{4.2}$$

Here f is a marginal density of the joint distributions

$$(X, U) \sim \mathcal{U}\{(x, u) : 0 < u < f(x)\}. \tag{4.3}$$

By generating uniform random variables on the set $\{(x,u) : 0 < u < f(x)\}$ we can generate from our distribution in (4.2). Moreover, as f is the marginal distribution of X, we therefore generated a random variable from f. This generation in turn only used an evaluation of $f(x)$. This means that simulating $X \sim f(x)$ is equivalent to simulating

$$(X,U) \sim \mathcal{U}\{(x,u) : 0 < u < f(x)\}. \tag{4.4}$$

In order to get close to the distribution of f, one can simulate an entire pair (X,U) in a bigger set, where simulation is possible and then select the pairs that satisfy the constraints for the desired smaller set and reject the pairs that do not. Considering a one dimensional setting with

$$\int_a^b f(x)\mathrm{d}x = 1, \tag{4.5}$$

where f is bounded by a constant m. By simulating $Y \sim \mathcal{U}(a,b)$ and then $U|Y = y \sim \mathcal{U}(0,m)$ we get a proposed pair. This pair is accepted only if $0 < u < f(y)$. X being the distribution of the accepted values for Y then has $f(x)$ as density function. As the distribution of X is

$$\begin{aligned} P(X \leq x) = \frac{P(Y \leq x \cap U < f(y))}{P(U < f(y))} &= \frac{\int_a^x \int_0^{f(y)} \frac{1}{m}\mathrm{d}u\mathrm{d}y}{\int_a^b \int_0^{f(y)} \frac{1}{m}\mathrm{d}u\mathrm{d}y} \\ &= \frac{m}{m}\frac{\int_a^x \int_0^{f(y)} \mathrm{d}u\mathrm{d}y}{\int_a^b \int_0^{f(y)} \mathrm{d}u\mathrm{d}y} \overset{(4.2)}{=} \frac{\int_a^x f(y)\mathrm{d}y}{\int_a^b f(y)\mathrm{d}y} \overset{(4.5)}{=} \int_a^x f(y)\mathrm{d}y. \end{aligned} \tag{4.6}$$

So far we assumed there is a constant m for which $f(x)$ is bounded for all $x \in \mathbb{R}$ and we only sampled from an interval $[a,b]$ that only spans a subset of the domain $\mathbb{R}$. A potentially better choice for a larger set, where on the one hand (i) sampling from is feasible and that (ii) acts as an upper bound for our $f(x)$ is the scaling of another density r s.t. $Mr(x) \geq f(x)$ for all $x \in X$, where X is a general probability space. $Mr(x)$ acts as an envelope for the target density. Figure 4 illustrates an example target density and a corresponding envelope.

Analogue to the above approach, to avoid the difficulty of sampling from f or the equally difficult task of sampling uniformly under the graph of $f(x)$ we could sample uniformly under the graph of the envelope $Mr(x)$ and only accept the samples that also fall under the graph of $f(x)$. For that it is important that the envelope is chosen in such a way that sampling from it is feasible.
Algorithmically we start by sampling an x from the density r. We then compute $f(x)$

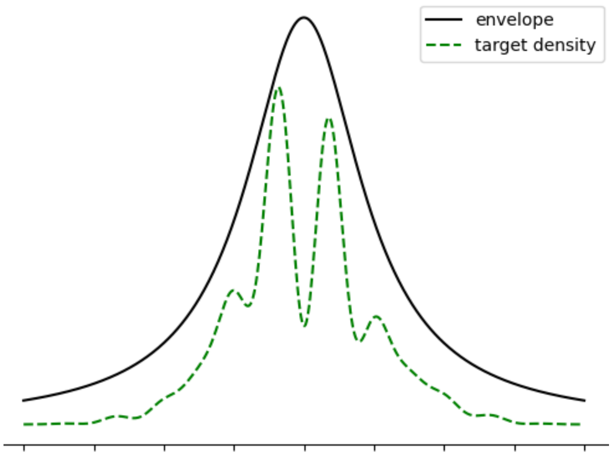

Figure 4: Illustration of the target density and the corresponding envelope function.

and $Mr(x)$. Independently of x, a uniform random variable U is generated. If $UMr(x) \leq f(x)$ holds, x and U are accepted, if not, they are discarded. The mentioned steps are shown in Algorithm 1.

Algorithm 1 Accept-Reject algorithm

Repeat: Generate the two random variables $\xi \sim r$ and $U \sim U([0, 1])$
if $U \leq f(\xi)/(Mr(\xi))$ **then**
  Accept (ξ, U)
else
  Reject (ξ, U)
end if

To verify that this algorithm is correct and that the accepted pairs (ξ, U) are in fact distributed according to our target density f, follows from the following two propositions. Both results are influenced by Proposition 6.2.2 and Lemma 6.2.3 in the book by Cappé, Moulines and Rydén [Cappe et al., 2005].

Proposition 4.1. *Let ξ be a random variable with f as a probability density w.r.t. to a measure λ on X and let U be a real random variable uniformly distributed on the interval $[0, M]$. This random variable U is independent from ξ.*
Then (i) the pair $(\xi, Uf(\xi))$ of random variables is then uniformly distributed on $G_{f,M} = \{(x, u) \in X \times \mathbb{R} : 0 < u < Mf(x)\}$, w.r.t. to $\lambda \otimes \lambda^{Leb}$, where λ^{Leb} denotes the Lebesgue measure.
Conversely, (ii) if a random vector (ξ, U) on $X \times \mathbb{R}$ is uniformly distributed on $G_{f,M}$,

then ξ admits f as its marginal probability density function.

Proof. Without loss of generality we take $M = 1$. If the proposition holds for an M_0, the statements also hold for all $M > 0$ by rescaling the graph by the factor M/M_0.
For statement (i), consider a measurable subset $B \subset G_{f,1}$ and B_x, the section of B given a fixed x, i.e. $B_x = \{u : (x,u) \in B\}$. As $Uf(x) \sim U([0, f(x)])$, $Uf(x)$ has the density $1/f(x)$ and ξ the density $f(x)$. Therefore, we have

$$P\{(\xi, Uf(\xi)) \in B\} = \int_{x\in X} \int_{u\in B_x} \frac{1}{f(x)} \lambda^{Leb}(\mathrm{d}u) f(x) \lambda(\mathrm{d}x)$$
$$= \int_B (\lambda^{Leb} \otimes \lambda)(\mathrm{d}x, \mathrm{d}u) = \int_B 1(\lambda^{Leb} \otimes \lambda)(\mathrm{d}x, \mathrm{d}u).$$

Thus, $(\xi, Uf(\xi))$ is uniformly distributed on $G_{f,1}$.

For statement (ii), consider a measurable subset $A \subset X$ and set
$\bar{A} = \{(x,u) \in A \times \mathbb{R} : 0 \leq u \leq f(x)\}$ the graph of f for all $x \in A$. In addition set $\bar{G}_x = \{(u : (x,u) \in_{f,1}\}$. Then

$$P(\xi \in A) = P((\xi, U) \in \bar{A}) = \frac{\int \int_{\bar{A}} \lambda^{Leb}(\mathrm{d}u)\lambda(\mathrm{d}x)}{\int \int_{G_{f,1}} \lambda^{Leb}(\mathrm{d}u)\lambda(\mathrm{d}x)} =$$
$$\frac{\int_{x\in A} \int_{u\in G_x} 1/f(x) \lambda^{Leb}(\mathrm{d}u) f(x)(\mathrm{d}x)}{\int_{x\in X} \int_{u\in G_x} 1/f(x) \lambda^{Leb}(\mathrm{d}u) f(x)(\mathrm{d}x)} = \int_A f(x)\lambda(\mathrm{d}x).$$

Where the last step is true as $\int_{u\in G_x} 1/f(x)^{Leb}(\mathrm{d}u)$ cancels out and $\int_{x\in X} f(x)\lambda(\mathrm{d}x) = 1$. ■

Proposition 4.2. *Let $V_1, V_2, \ldots$ be a sequence of i.i.d. random variables the measurable space $(V, \mathcal{V})$ and let $B \in \mathcal{V}$ a set such that $P(V_1 \in B) = p > 0$.*
(i) The random variable $\sigma = inf\{k \geq 1, V_k \in B\}$ (with $inf\{\emptyset\} = \infty$) is geometrically distributed with parameter p, i.e., for all $i \geq 0$,

$$P(\sigma = i) = (1-p)^{i-1}p.$$

(ii) The random variable $V = V_\sigma \mathbf{1}_{\sigma<\infty}$ is distributed according to

$$P(V \in A) = \tfrac{P(V \in A \cap B)}{p}.$$

Proof. To prove (i) we use the independence of the $V_1, V_2 \ldots$ and compute $P(\sigma = i) = P(V_1 \notin B, \ldots, V_{i-1} \notin B, V_i \in B) = (1-p)^{i-1}p$. This is exactly the discrete probability

distribution of the geometric distribution. This also implies that the waiting time σ is finite with probability one, as $p > 0$.
To prove (ii) consider a set $A \in \mathcal{V}$ and

$$P(V \in A) = \sum_{i=1}^{\infty} P(V_1 \notin B, ..., V_{i-1} \notin B, V_i \in A \cap B)$$
$$= \sum_{i=1}^{\infty}(1-p)^{i-1} P(V_1 \in A \cap B) = P(V_1 \in A \cap B)\frac{1}{1-(1-p)} = \frac{P(V_1 \in A \cap B)}{p}.$$

Where in the second step, we again used that the $V_1, V_2, ...$ are i.i.d. and that for each of the V_i it holds that $P(V_i \in B) = p$. After that we used that $\sum_{i=1}^{\infty}(1-p)^{i-1}$ is the geometric series. ■

By in Proposition 4.1 (i), the intermediate (not necessarily accepted) pair (ξ_i, U_i) generated in the algorithm is such that $(\xi_i, MU_i r(\xi))$ are uniformly distributed under the graph of $Mr(x)$. Note that in the algorithm we sample $U \sim U([0,1])$ and not $U \sim U([0,M)]$. Therefore, $(\xi_i, MU_i r(\xi))$ is uniform under $Mr(x)$ and not $(\xi_i, U_i r(\xi))$.
By Proposition 4.2 (ii) the accepted pair (ξ, U) is then uniformly distributed under the graph of $f(x)$. The variables V_i are the samples that are drawn from the envelope $Mr(x)$, V is an accepted sample, A is the bigger set where sampling from is feasible and B is the set we want to sample from.
By Proposition 4.1 (ii) ξ then admits f as marginal density. In addition, the acceptance probability p from Proposition 4.2 is:

$$P(U_1 \leq \frac{f(\xi_1)}{Mr(\xi_1)}) = P((\xi_1, MU_1 r(\xi_1)) \in G_{f,M}) = \frac{\int_X f(x)\lambda(\mathrm{d}x)}{\int_X Mr(x)\lambda(\mathrm{d}x)} = \frac{1}{M}.$$

The algorithm is therefore correct and gives us a way to approximate f only having access to $f(x)$. One thing to consider when choosing r and M is that $f(x) \leq Mr(x)$ is not given for all $x \in X$, if the density r does not have heavier tails as well as sharper infinite peaks than the target distribution f. For example, as shown in Figure 5, sampling from a student's t distribution with a normal distribution as envelope yields an approximation of f that is not accurate.

A student's t distribution, however would be an appropriate choice in order to i.i.d. sample from a normal distribution as can be seen in Figure 6.

The algorithm is more efficient, the fewer pairs of (ξ, U) get rejected, so it is essential to keep the area between the graphs as small as possible. This area in one dimension equals $\frac{1}{M}$. The difficulty for a reliable and efficient algorithm thus lies in the choice of proposal density r and choosing M_r as small as possible.
The task of finding a proposal r and a constant M_r turns out to be particularly hard in

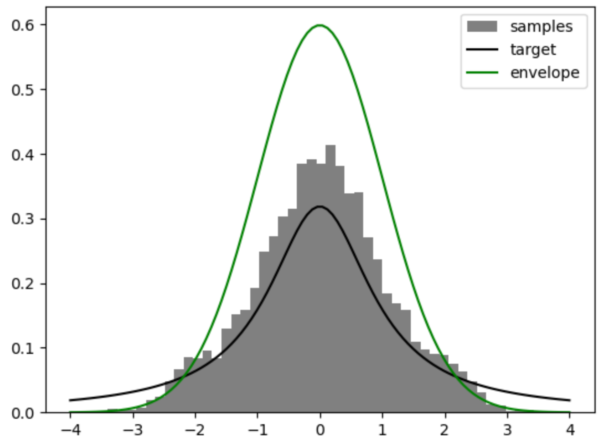

Figure 5: Normal distribution density with a scaling factor of $M = 1.5$ as envelope for a student's distribution density. The histogram shows the approximation after 10,000 steps of the Accept-Reject algorithm.

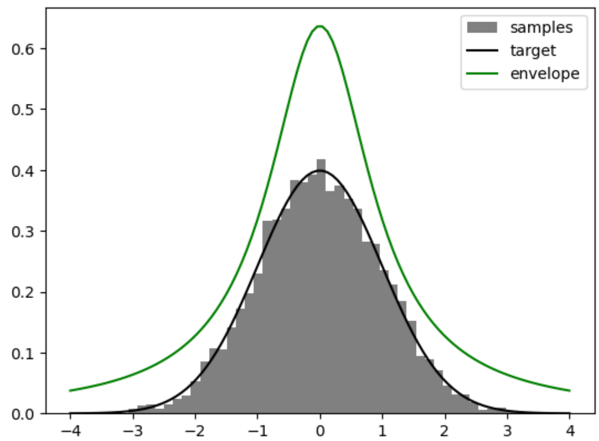

Figure 6: Student distribution density with a scaling factor of $M = 2$ as envelope for a normal distribution density. The histogram shows the approximation after 10,000 steps of the Accept-Reject algorithm.

high dimensional settings. The following example illustrates how the acceptance probability shrinks with the dimension.
Splitting $\pi(x)$ as well as $r(x)$ in a constant part and a functional part yields $\bar{\pi}(x)/C_\pi$ and $\bar{r}(x)/C_r$. Where C_π and C_r are constants and $\bar{\pi}(x)$ and $\bar{r}(x)$ are the functional parts. As

$$\pi(x)/r(x) \leq M \Leftrightarrow \bar{\pi}(x)/\bar{r} \leq \bar{M} \; with \; \bar{M} = C_\pi M/C_r$$

one can ignore the constants and instead find an $\bar{M}$ that bounds $\bar{\pi}(x)/\bar{r}$.
Considering two normal distribution densities, one with variance 1 and the other with variance σ^2 we obtain

$$\bar{\pi}(x_1,...,x_d) = exp(-\tfrac{1}{2}\textstyle\sum_{k=1}^{d} x_k^2) \ with\ C_\pi = (2\pi)^{d/2} \text{ and}$$
$$\bar{r}(x_1,...,x_d) = exp(-\tfrac{1}{2\sigma^2}\textstyle\sum_{k=1}^{d} x_k^2) \ with\ C_r = (2\pi\sigma^2)^{d/2}$$

For a $\sigma > 1$ the ratio can be bounded by $\bar{M} = 1$ as

$$\frac{\bar{\pi}(x_1,...,x_d)}{\bar{r}(x_1,...,x_d)} = exp(-\frac{1}{2}(1-\sigma^{-2})\sum_{k=1}^{d} x_k^2) \leq 1 = \bar{M}$$

The acceptance probability is then

$$P(U \leq \frac{\bar{\pi}(X_1,...,X_d)}{\bar{M}\bar{r}(X_1,...X_d)}) = \frac{C_\pi}{\bar{M}C_r} = \frac{(2\pi)^{d/2}}{(2\pi\sigma^2)^{d/2}} = \sigma^{-d}$$

Thus, the acceptance probability shrinks exponentially with higher dimensions and makes the algorithm potentially inefficient in high dimensions. This phenomenon is called the *curse of dimensionality*. To avoid inefficient simulations in cases we will explore methods that are less susceptible by the dimensionality.

4.2 The Metropolis Hastings Algorithm

An alternative to sampling from arbitrary distributions in large-dimensional settings are simulation methods called *Markov chain Monte Carlo (MCMC)* methods. Note that the "Markov" in "Markov chain Monte Carlo" has nothing to do with the "Markov" in "hidden Markov models". MCMC methods can be a useful tool in HMM scenarios, but have a more general applicability that goes further than our focus in this thesis. MCMC approaches take advantage of the following principle. Sampling an i.i.d. sequence $\xi^1, ..., \xi^n$ with a common probability distribution π one can approximate π. If one considers a sequence $\xi^1, ..., \xi^n$ generated from a Markov chain the ergodic theorem for Markov chains states that

$$\hat{\pi}_N^{MCMC}(f) := \frac{1}{N} \sum_{i=1}^{N} f(\xi^i) \tag{4.7}$$

is an estimate of the expectation of f under the stationary distribution of the chain $\{\xi^i\}_{i \geq 1}$, for all integrable functions f.
For further details on the ergodic theorem of Markov chains we refer to chapter 14 in the book by Cappé, Moulines and Rydén [Cappe et al., 2005].

In order to apply the result in (4.7) there are some requirements for the simulation to be successful:
(I) simulating $\{\xi^i\}_{i \geq 1}$ given an arbitrary ξ^1 is implementable
(II) the stationary distribution of $\{\xi^i\}_{i \geq 1}$ coincides with the target distribution π
(III) the chain $\{\xi^i\}_{i \geq 1}$ satisfies some conditions to guarantee the convergence towards π irrespective of the initial ξ^1
The following algorithm was first introduced in 1953 [Metropolis et al., 1953] and then later generalised in 1970 [Hastings, 1970]. We can simulate a sequence of values $\{\xi^i\}_{i \geq 1}$ forming a Markov chain on X. Choosing an initial ξ^1 arbitrarily and given a ξ^i we generate the next value with the following algorithm, where r is a transition density function called proposal distribution, whose functional form is known:

Algorithm 2 Metropolis-Hastings algorithm

Generate $\xi \sim r(\xi^i, \cdot)$

Set $\xi^{i+1} = \begin{cases} \xi & \textit{with probability } \alpha(\xi^i, \xi) \stackrel{\text{def}}{=} \frac{\pi(\xi) r(\xi, \xi^i)}{\pi(\xi^i) r(\xi^i, \xi)} \wedge 1 \\ \xi^i & \textit{otherwise.} \end{cases}$

Regarding (II), the acceptance probability $\alpha(\xi^i, \xi)$ is chosen in such a way that the associated Markov chain $\{\xi_t\}$ satisfies the detailed balance equation, thus making the transition kernel associated to the algorithm reversible. In addition, π is the chains stationary distribution.

Proposition 4.3 (Reversibility of the Metropolis-Hastings Kernel)**.** *The chain* $\{\xi^1\}_{i\geq 1}$ *that is generated by Algorithm 2 has the following properties:*
(i) the chain is reversible, and (ii) π *is the chains stationary probability density function.*

Proof. For (i) it is sufficient to show that the Kernel associated to the Metropolis Hastings algorithm fulfills the detailed balance equation with the density π.
The Kernel K associated to the algorithm is of the following form:
$K(x, x') = \alpha(x, x')r(x, x') + p(x)\delta_x(x')$ where $p(x) = 1 - \int_X \alpha(x, x')r(x, x')\lambda(\mathrm{d}x')$. The first part of the sum corresponds to the probability of transitioning from x to x' with acceptance probability α. The second part of the sum corresponds to the probability of staying at x in case $x' = x$, where $p(x)$ is the complementary probability of the chain moving from x to any other $x' \neq x$.
It follows that

$$\begin{aligned} K(x, x')\pi(x) &= (\alpha(x, x')r(x, x') + p(x)\delta_x(x'))\pi(x) \\ &= \left(\frac{\pi(x)r(x', x)}{\pi(x)r(x, x')} \wedge 1\right) r(x, x')\pi(x) + p(x)\delta_x(x')\pi(x) \end{aligned}$$

where $p(x)\pi(x)$ is obviously symmetric as it only applies if $x = x'$ and for the first part of the sum

$$\left(\frac{\pi(x)r(x', x)}{\pi(x)r(x, x')} \wedge 1\right) r(x, x')\pi(x) = \pi(x')r(x', x) \wedge \pi(x)r(x, x') = \pi(x)r(x, x') \wedge \pi(x')r(x', x)$$

holds, as the minimum function is symmetric.
Therefore, K satisfies the detailed balance equation and is therefore reversible and (ii) follows from theorem 3.6. ■

This result however, does not guarantee requirement (III), that the chain $\{\xi^i\}_{i\leq 1}$ actually converges to π in distribution, independent of the choice of ξ^1. For these requirements we refer to chapter 14 in [Cappe et al., 2005].

If we look at the acceptance probability $\alpha(\xi^i, \xi)$ and we leave out the proposal r for a moment, the acceptance is higher if $\pi(\xi) > \pi(\xi^i)$. Thus, accepting values for ξ is more likely in a region with more mass and less likely in a region with less mass. A visualisation of this concept is shown in Figure 7. Starting with $\xi = 1$ the probability that a step to $\xi = -0.5$ will be accepted is much higher than that a step to $\xi = 2$ will be accepted.

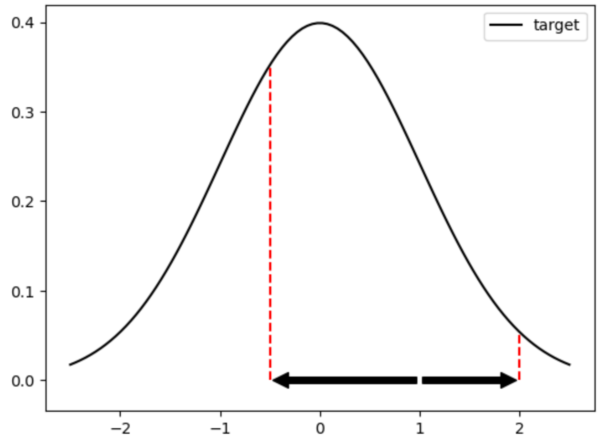

Figure 7: Visualisation of two proposed steps in the Metropolis-Hastings algorithm. The arrows are possible steps the algorithm could take. The red dotted line is the corresponding $\pi(\xi)$ value for the step.

One important feature of the Metropolis-Hastings algorithm that makes it viable for HMM's is that it can be applied when π and r are only known through the ratio $\pi(x')/\pi(x)$ or $r(x', x)/r(x, x')$, thus not needing normalising constants. Only evaluating π and r up to a constant scale factor or only knowing π/r is sufficient to apply the algorithm. Indeed, in our case this feature is helpful for simulating from the posterior distribution of an unobservable sequence of states $X_{0:n}$ given observations $Y_{0:n}$ as given in equation (3.5). The functional form is fully explicit except for the normalisation factor $L_{\nu,n}^{-1}$. In MCMC approaches this factor can be neglected as for working with a ratio, it suffices that the joint target distribution is proportional to

$$\phi_{0:n|n}(y_{0:n}, x_{0:n}) \propto \nu(x_0)g_0(x_0, y_0) \prod_{k=1}^{n} q(x_{k-1}, x_k)g_k(x_k, y_k) \tag{4.8}$$

where the model is dominated by λ and ν and q are a density function and a transition density function.

4.2.1 Independent Metropolis-Hastings

An easy choice for the proposal transition density function $r(x, \cdot)$ is to choose a distribution over X that is independent of x. The acceptance probability then reduces to

$$\alpha(x, x') = \frac{\pi(x')/r_{ind}(x')}{\pi(x)/r_{ind}(x)} \wedge 1$$

As Mengerson and Tweedie [Mengersen and Tweedie, 1996] showed, the performance of an independent Metropolis-Hastings algorithm depends on whether the importance ratio $\pi(\xi)/r_{ind}(\xi)$ is bounded. For example, the Cauchy(0,1) target distribution and a Gaussian N(0,1) proposal distribution have a ratio of

$$\pi(x)/r_{ind}(x) = exp(x^2/2)/(1 + x^2),$$

which is unbounded. Similarly to the Accept-Reject algorithm the Gaussian proposal distribution cannot reach the tails of the Cauchy distribution as seen in Figure 8.

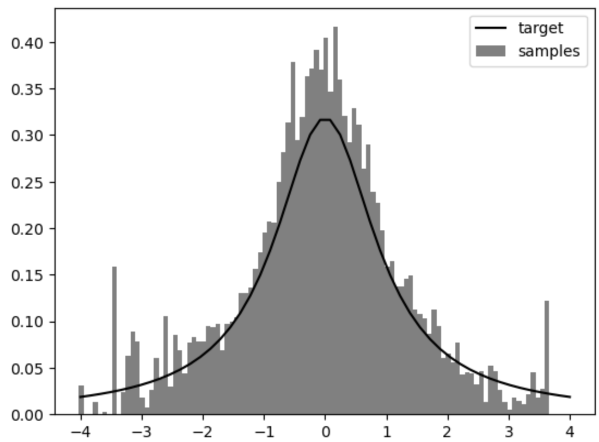

Figure 8: Histogram of the independent Metropolis-Hastings algorithm of a Cauchy distribution approximated with a normal distribution as proposal with a chain length of 10,000 steps.

An example, where the approximation yields better results, is shown in Figure 9, where a uniform proposal distribution is used for a Gaussian target distribution. Using a uniform proposal distribution reduces the importance ratio to $\pi(x)/r_{ind}(x) = \pi(x) = (2\pi)^{-1/2}exp(-\frac{1}{2}x^2)$ (as $r_{ind}((x')) = r_{ind}(x)$), which is of course bounded.

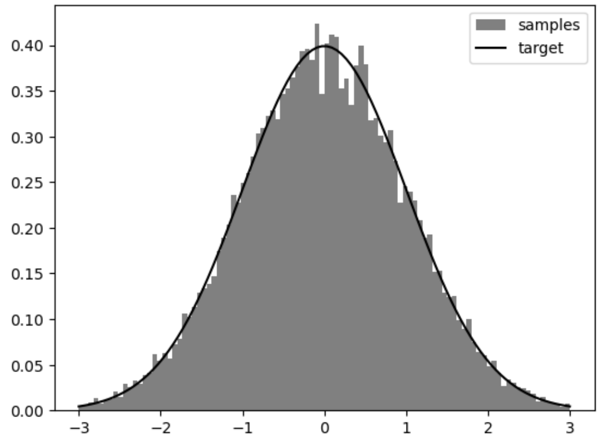

Figure 9: Histogram of the independent Metropolis-Hastings algorithm of a normal distribution approximated with a uniform distribution as proposal with a chain length of 50,000 steps.

4.2.2 Random Walk Metropolis-Hastings

Another option for the proposal distribution $r(x,\cdot)$ is some sort of local movement around x. On a continuous state space one could use a random walk on X such as $r(x,x') = h(x'-x)$, where h is symmetric as proposal density. As h is symmetric the acceptance probability, again, reduces to $\alpha(x,x') = \frac{\pi(x')}{\pi(x)}$. In Figure 10, a Student distribution with 1 degree of freedom is approximated with an N(0,5) random walk as proposal distribution. Even though Figure 10 shows a good approximation, there were some choices made to ensure the adequacy of the approximation. The two main ways in which the algorithm may still yield bad results are the tail behaviour and the scale or variance of the random walk.

An example for different approximation performances can be seen when looking at bimodal distributions. The target distribution in Figure 11 and 12 is based on two Gaussian distributions with different means and scales. Consider a density f of the distribution $X \sim N(0,1)$ and a density g of the distribution $Y \sim N(6,0.5)$. A new probability density $q(x) = \frac{1}{2}(f(x)+g(x))$ is then the density of a bimodal distribution. Figure 11 shows the histogram of a chain after 10,000 steps. As the scale of the random walk is only 0.2, the chain never jumps over to the other mode. The overestimation of one mode is due to the fact that histogram is normalised to portray a density and thus attributes all the mass at the only explored mode. Thus, in absence of the exploration of a substantial part of the mass of the distribution, the approximation seems like an N(0,1) approximation. In Figure 12 the scale of the random walk is 5, leading to the chain jumping between the

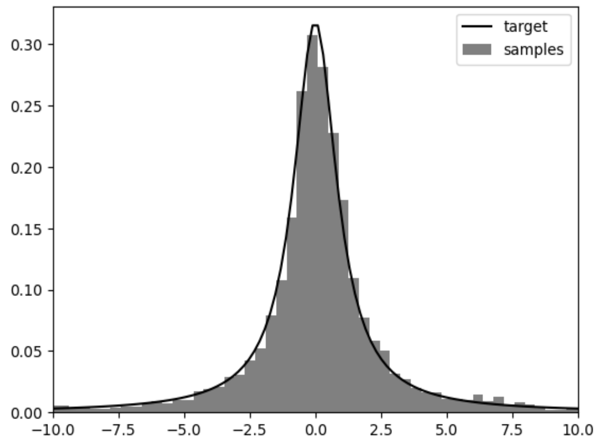

Figure 10: Histogram of the random walk Metropolis-Hastings algorithm of a student distribution with 1 degree of freedom approximated with a normal distribution as proposal with a chain length of 10,000 steps.

modes several times, therefore approximating the density well.

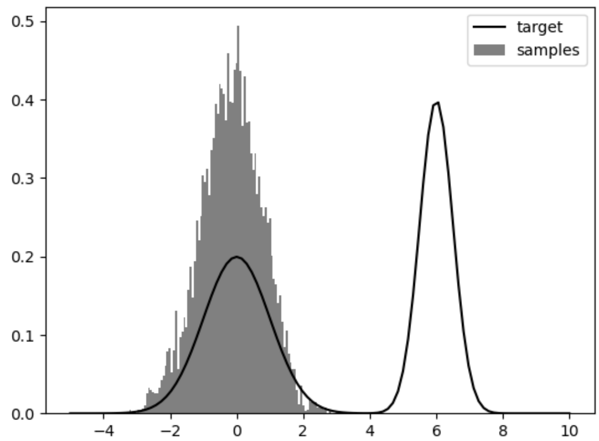

Figure 11: Histogram of a bimodal distribution approximated with a normal random walk proposal distribution with low variance (N(0,0.2)) with a chain length of 10,000 steps.

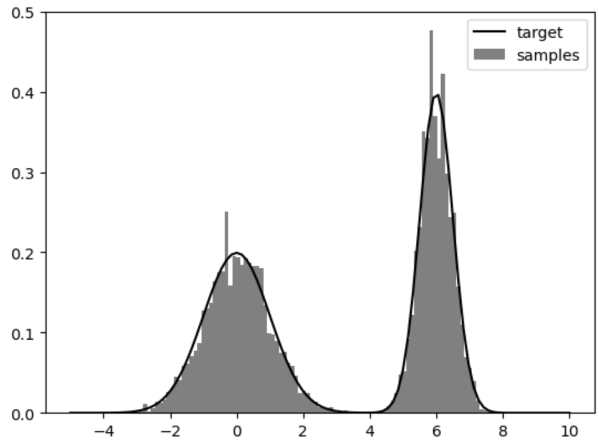

Figure 12: Histogram of a bimodal distribution approximated with a normal random walk proposal distribution with high variance (N(0,5)) with a chain length of 10,000 steps.

4.3 Gibbs Sampling

In a multivariate scenario the conditional distribution of a single variable given all remaining variables may have a form that is simple enough to be evaluated. As mentioned earlier in the introduction, HMM's have a very specific conditional independence structure. The observable process Y_k only depends on X_k and is independent of all other hidden states. Thus in HMM scenarios, a natural choice for an MCMC algorithm is the *Gibbs Sampler*. The name Gibbs sampler stems from the fact that a paper by Geman and Geman highlighted its usefulness for the simulation of so called Gibbs Markov random fields [Geman and Geman, 1984]. Consider the case where we are given a joint probability density function π on a space X where x is decomposable in $x = (x_1, ..., x_m)$ with $x_k \in X_k$. In addition let $x_{-k} = x_{l \neq k}$ be the notation for all components except for x_k. Analogue to this notation, the conditional probability density function of X_k given $\{X_l\}_{l \neg k}$ is given by $\pi(\cdot|x_{-k})$. Assuming that simulation from this distribution is feasible, consider the following algorithm:

Algorithm 3 Gibbs sampler

Starting with an arbitrary ξ^1, the update rule for the current state $\xi^i = (\xi^i_1, ..., \xi^i_m)$ to the next state ξ^{i+1} is as follows:
for k=0 to m **do**
  Simulate ξ^{i+1}_k from $\pi_i(\cdot|\xi^{i+1}_i, ..., \xi^{i+1}_{k-1}, \xi^i_{k+1}, ..., \xi^i_m)$.
end for

This means that the kth round of the simulation of ξ^{i+1}, the kth component is updated by simulating from the conditional distribution of all other components. In other words, the complete cycle can be seen as a combination of m single MCMC steps where in each step one of the components is being modified. Also, each of the the individual m steps in Algorithm 3 is π reversible and thus admits π as a stationary probability density function. Proofing this involves a similar argument as the proof of Proposition 4.3 and relies on showing that the detailed balance equation holds for each of the individual steps. For the covering of convergence conditions we refer to chapter 8.3 in the book by Robert and Casella [Robert and Casella, 2004].

Looking back to our description of the Accept-Reject Algorithm, in Proposition 4.1 we showed that if the random variable (ξ, U) on $X \times \mathcal{R}$ is uniformly distributed on $G_f = \{(x, u) \in X \times \mathcal{R}^+ : 0 \leq u \leq f(x)\}$, then the marginal distribution on X is f. As a new notation we will use U not only as uniform distribution on an interval, but also on sets, e.g. $U(G_f)$ would then be the uniform distribution on the set G_f. Designing an MCMC algorithm we could use a random walk on this set G_f to produce a Markov chain with a stationary distribution that coincides with the uniform distribution on G_f. Keeping the idea of Algorithm 3 in mind, one could construct a random walk on G_f in going in one direction at a time, first moving along the u-axis and then along the x-axis. Both moves are at random and more specifically we can choose our step from a uniform distribution. Starting from a point (x, u) in G_f moving along the u-axis corresponds to the conditional distribution $U(\{u : u \leq f(\xi)\})$ jumping from point (x, u) to point (x, u'). Moving along the x-axis corresponds to the conditional distribution $U\{x : f(x) \geq u'\}$ jumping from (x, u') to (x', u').

Algorithm 4 Slice Sampler

Starting from an arbitrary initial point (ξ^1, U^1) in G_f.
for $i \geq 1$ **do**
  $U^{i+1} \sim U([0, f(\xi^i)])$
  $\xi^{i+1} \sim U(S(U^{i+1}))$, with $S(u) = \{x : f(x) \geq u\}$.
end for

The slice sampler algorithm is a Gibbs sampling method in the sense that both steps are conditional distributions of U and ξ of the joint distribution of (ξ, U) on $U(G_f)$.
As illustration of the above described process, the first 10 steps of the slice sampler approximating a normal distribution as target are shown in Figure 13.

To apply the slice sampler we will now come back to the earlier introduced stochastic

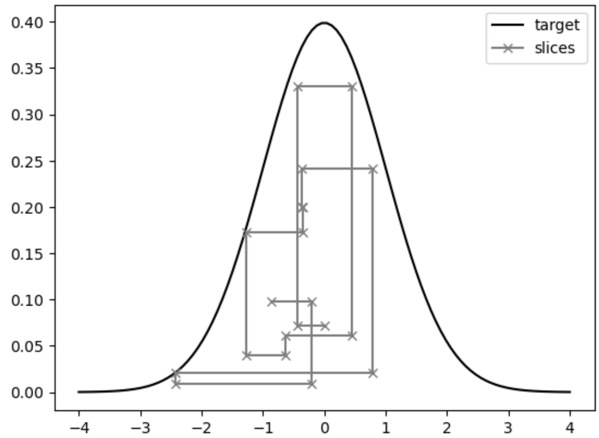

Figure 13: Visualisation of the first 10 steps of the slice sampler for a normal distribution as target.

volatility model that looks as follow:

$$
\begin{aligned}
X_{k+1} &= \phi X_k + \sigma U_k, \qquad U_k \sim N(0,1) \\
\mathrm{Y}_k &= \beta\ \exp(X_k/2) V_k, \qquad V_k \sim N(0,1).
\end{aligned}
$$

This example is at its core analogue to example 6.2.16 in the book by Cappé, Moulines and Rydén [Cappe et al., 2005]. Our goal is to approximate the latent process $\{X_k\}_{k\geq 0}$: the log-volatility. As given in equation (4.8) the joint conditional distribution of $X_{0:n}$in given $Y_{0:n}$ up to constants is given by $\phi_{0:n|n}(y_{0:n}, x_{0:n}) \propto \nu(x_0) g_0(x_0, y_0) \prod_{k=1}^{n} q(x_{k-1} x_k) g_k(x_k, y_k)$. The conditional probability density of a single variable X_k in the hidden chain, given $Y_{0:n}$ and the two neighbours X_{k-1} and X_{k+1} is such that

$$
\phi_{k-1:k+1|n}(y_k, x_k | x_{k-1}, x_{k+1}) \propto \phi_{k-1:k+1|n}(y_k, x_{k-1:k+1}) \propto q(x_{k-1}, x_k) q(x_k, x_{k+1}) g_k(x_k, y_k) \tag{4.9}
$$

So simulating $X_{0:n}$ one component at a time, we have the right expression in equation (4.9). Rearranging the model definition we get:

$$
\begin{aligned}
q(x_{k-1}, x_k | x_{k-1}) &\sim N(\phi X_k, \sigma^2), \\
q(x_k, x_{k+1} | x_{k+1}) &\sim N(\phi X_{k-1}, \sigma^2), \\
g(x_k, y_k | y_k) &\sim N(0, \beta^2 \mathrm{exp}(X_k)).
\end{aligned}
$$

Thus, leaving out constants we get:

$$
\exp\left[-\left\{ \frac{(x_{k+1} - \phi x)^2}{2\sigma^2} + \frac{(x - \phi x_{k-1})^2}{2\sigma^2} \right\} \right] \frac{1}{\beta \mathrm{exp}(x/2)} \mathrm{exp}\left[-\frac{y_k^2}{2\beta^2 \mathrm{exp}(x)} \right].
$$

Completing the square in x, we can rearrange the density to

$$\exp\left[-\frac{1+\phi^2}{2\sigma^2}\left\{(x-\mu_k)^2+\frac{y_k^2\sigma^2}{(1+\phi^2)\beta^2}\exp(-x)\right\}\right],$$

where

$$\mu_k=\frac{\phi(x_{k+1}+x_{k-1})-\sigma^2/2}{1+\phi^2}.$$

If we also define

$$\alpha_k=\frac{y_k^2\sigma^2\exp(-\mu_k)}{(1+\phi^2)\beta^2}\quad\text{and}\quad\rho=\frac{1+\phi^2}{2\sigma^2},$$

we get the simpler expression

$$\exp\left[-\rho\left\{(x-\mu_k)^2+\alpha_k\ \exp[-(x-\mu_k)]\right\}\right]. \tag{4.10}$$

As μ_k is a shift that does not affect the simulation, the final form we are going to use in the slice sampler is $\exp[-\rho\{x^2+\alpha_k\ \exp(-x)\}]$. In a data driven scenario one would use values fitted to past data for the parameters β, ϕ and σ. With these we have μ_k, α_k and ρ and equation (4.10) can be used for the part in the slice sampler where we sample $\xi^{i+1} \sim U(S(U^{i+1}))$ with $S(u) = \left\{x : \exp\left[-\rho\left\{x^2+\alpha_k\ \exp(-x)\right\}\right] \geq u\right\}$. An example approximation of the distribution of X_k with a slice sampler for the values $\rho = 1$ and $\alpha_k = 5$ is given in Figure 14.

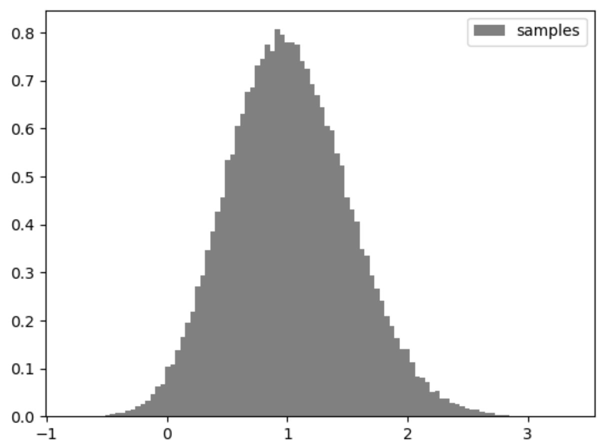

Figure 14: Histogram of the distribution of X_k approximated with the slice sampler using 100,000 steps.

5 Conclusion

We have looked at several algorithms that enable sampling from probability distributions. The first proposed algorithm was the Accept-Reject algorithm. We have proven that the algorithm works and have seen that the choice of proposal density has an influence on the approximation. In addition, we have shown that this algorithm becomes more computationally expensive in high-dimensional settings. The second and third algorithm fell under the category of Markov chain Monte Carlo methods, as we used the stationary distribution of a Markov chain to approximate our target distribution. These algorithms have the advantage that we can neglect any constant normalisation factors as these cancel out. More specifically we have seen that the Likelihood term does not need to be considered. The Metropolis-Hastings algorithm generates a chain that is reversible and the stationary density is in fact the one we are looking for. Two different choices of proposal distributions namely an independent proposal distribution and a random walk proposal distribution served as examples for some numerical experiments. We then went on to look at the slice sampler, which makes use of the Gibbs sampling principle. With the help of the slice sampler we were able to approximate a hidden state distribution in an HMM example regarding stock price changes.

As stated before the Accept-Reject algorithm does not scale well in higher dimensions. However, even though the Metropolis-Hastings algorithm does not have this shortcoming, there are potential difficulties too. In the case of an independent proposal distribution for one, if the ratio between the proposal and target density is unbounded, the approximation may be inaccurate. In addition, in the case of random walk Metropolis-Hastings, the scale of the random walk steps plays a major role. As we have seen, in case of a bimodal distribution, or any distribution with clearly separated areas of high mass, small scale steps fail to identify all areas of high mass. Thus, in order to get a more reliable approximation of the target distribution one may need to try different starting values as well as different scales for the random walk. Regarding the slice sampler we did not conduct numerical experiments, in which the slice sampler fails to approximate the target distribution. Depending on the target distribution there can however, be an issue with the right choice for uniform sampling along the x-axis.

Apart from the fact that these algorithms have different advantages and disadvantages to consider, we have shown that MCMC methods prove useful in HMM contexts when making inferences on the distribution of hidden states.

The here presented algorithms lay a foundation on how use MCMC algorithms to make inferences in HMM scenarios. There are however, more options one could consider such as sequential Monte Carlo methods. Algorithms like the *sequential importance sampling* and its extension the *sequential importance sampling resampling* are interesting further options one might want to consider.

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